



Lab 12: VLSM and Dynamic Routing (RIP)

Variable Length Subnet Mask (VLSM)

In traditional IP subnetting, commonly referred to as Fixed Length Subnet Masking (FLSM), a single subnet mask is applied uniformly across an entire network. This means that every subnet created from the original network has the same number of available IP addresses, regardless of the actual number of hosts required in each segment. While this method is straightforward and easy to implement, it often results in inefficient utilization of IP address space. For instance, a subnet designed for only a few devices would still reserve the same number of addresses as a subnet intended for dozens of hosts, leading to a significant number of unused and wasted IP addresses.

Variable Length Subnet Masking (VLSM) addresses this inefficiency by allowing network administrators to create subnets of varying sizes within the same address space. Instead of applying a single fixed subnet mask, VLSM enables the division of a network into smaller sub networks based on specific host requirements. This method involves subnetting a network and then further subnetting the resulting subnets as needed. As a result, each subnet is allocated only as many IP addresses as it actually requires. For example, a subnet that supports 30 hosts can be assigned a larger block of addresses, while a point to point connection between two routers, which requires only two usable IP addresses, can be allocated a much smaller subnet. This flexible allocation significantly reduces IP address wastage and makes the overall network design more scalable and efficient, especially in large and complex network environments.

Dynamic Routing and RIP

Routing is a fundamental process in networking that determines how data packets travel from a source network to a destination network. In smaller or simpler network setups, routing decisions can be manually configured by administrators using static routes. While static routing provides full control over the path selection, it becomes increasingly difficult to manage as the network grows in size and complexity. Any change in the network topology, such as the addition of new networks or the failure of existing links, requires manual updates to routing tables, which can be time consuming and prone to human error.

Dynamic routing protocols were developed to overcome these limitations by automating the process of route discovery and maintenance. These protocols enable routers to communicate with each other, share information about reachable networks, and automatically update their routing tables based on changes in the network. This not only reduces administrative overhead but also improves the network's ability to adapt to failures and changes in topology.

The Routing Information Protocol (RIP) is one of the earliest and simplest dynamic routing protocols, classified as a distance vector protocol. It determines the best path to a destination based on a metric known as hop count, which represents the number of routers a packet must pass through to reach its destination. The path with the lowest hop count is considered the most



efficient. To prevent routing loops and excessive delays, RIP imposes a maximum limit of 15 hops, meaning that any destination beyond this limit is considered unreachable.

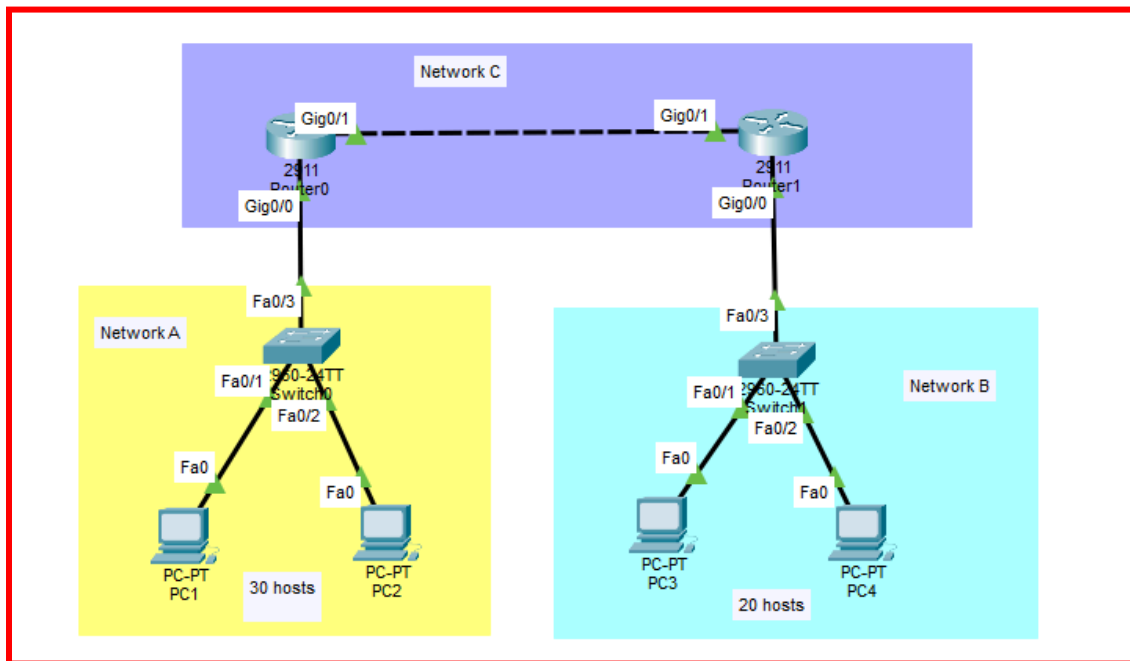
There are two types of RIP, RIP Version 1 (RIP v1) and RIP Version 2 (RIP v2). Unlike its predecessor, RIP v1, RIP v2 supports classless routing, which allows it to work with Variable Length Subnet Masks (VLSM). It also includes enhancements such as improved routing updates and better support for modern network designs.

Lab Tasks

Note: All tasks must be documented in a single report. Include screenshots of every step (description is not as necessary as the screenshots).

CLO2: Apply core networking concepts and protocols, including but not limited to routing, Congestion control, and error management, to establish and analyze reliable data communication across networks.

Task 1: VLSM Calculation and Implementation (10 marks)



You are given the base IP address block 190.120.12.0/24, and the above topology. The topology requires three separate networks. Network A, which needs 30 hosts. Network B, which needs 20 hosts and Network C which needs 2 hosts (because it's a point to point connection between two routers). Follow the steps provided to assign the required IP addresses.



Step 1: FLSM and VLSM Calculation

Watch the two provided instructional videos to understand the calculation method to find the appropriate IPs for the topology. The first [video will help you understand Fixed Length Subnet Mask \(FLSM\)](#). And the second video will [help you understand how to calculate the VLSM](#) based IP addresses needed.

Using the concepts and steps in the provided videos, calculate the required subnet masks, network addresses, usable IP ranges, and broadcast addresses for Networks A, B, and C using the VLSM method. Show your step by step calculations in your document and include a screenshot of your screen showing the instructional videos open, alongside a Notepad window containing your Name, Roll Number, Section, and "Lab 12" written on it.

Step 2: Setting Up the Network Topology and Assigning IPs

Open Cisco Packet Tracer and build the topology as per the provided diagram. Based on your VLSM calculations from Step 1, assign the appropriate IP configurations to your network components.

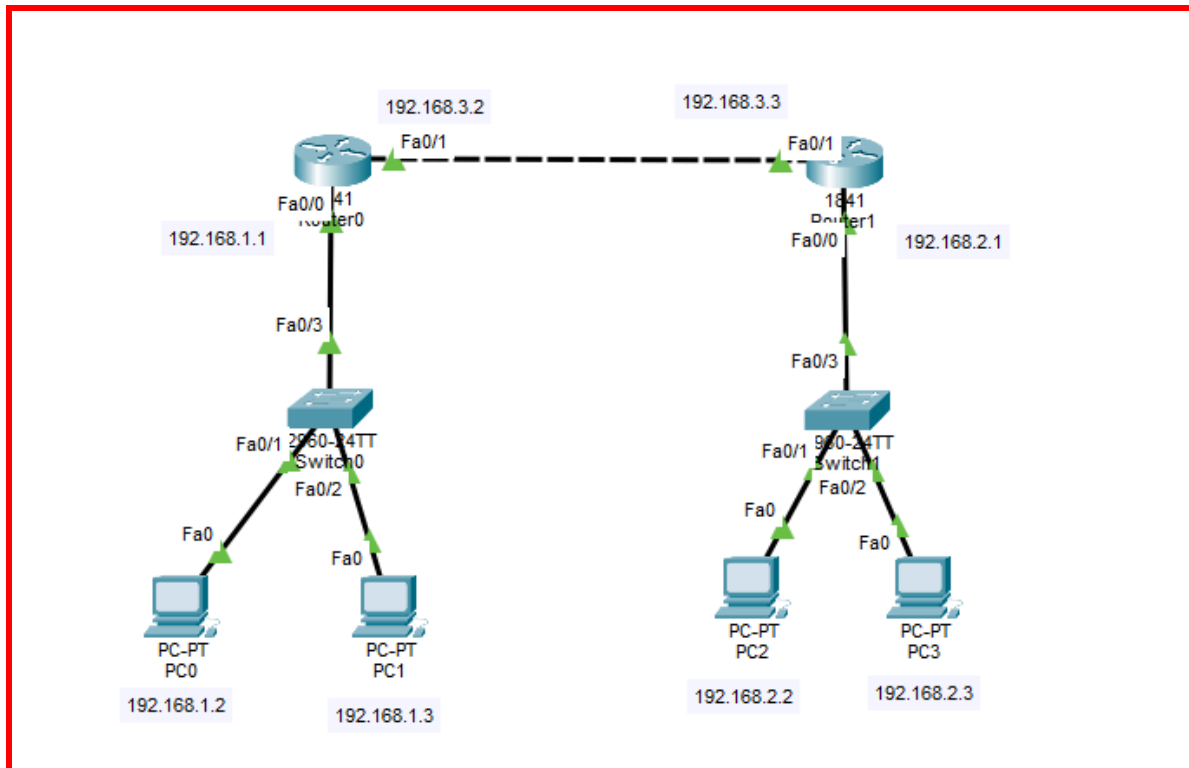
For the end devices (PCs): Navigate to Desktop > IP Configuration. Assign an appropriate static IP address, Subnet Mask, and Default Gateway to each PC in Network A and Network B.

Routers: Through the GUI or CLI, configure the router. You will need to configure two interfaces for each router (one GigabitEthernet interface connecting to the local LAN switch, and one GigabitEthernet interface connecting to the other router), with the appropriate IPs and subnet masks. You must also use the no shutdown command or turn on the port status for each router through the GUI.

Lastly, annotate the final topology appropriately with the corresponding network names, IPs, default gateways and subnet masks.



Task 2: RIP Implementation (10 marks)



Step 1: Setting Up the Topology and IP Configuration

Begin by opening Cisco Packet Tracer and constructing the required network topology. The topology consists of two 1841 routers, two switches (2960), and four PCs. Connect the devices such that each router is connected to one switch, and the two routers are interconnected. Each switch should have two PCs connected to it. Once the physical topology is complete, assign IP addresses to all devices according to the addressing scheme provided in the diagram. After assigning IP addresses, configure the default gateways on the PCs. For the PCs connected to the first network (PC0 and PC1), set the default gateway to 192.168.1.1, which corresponds to the interface of Router 0. For the PCs connected to the second network (PC2 and PC3), set the default gateway to 192.168.2.1, which corresponds to the interface of Router 1.

Step 2: Implementing RIP Routing Protocol

At this stage, the networks are isolated, devices on one network cannot communicate with devices on the other network. To enable communication between networks, you will configure the Routing Information Protocol (RIP) on both routers.



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For this purpose, go to Router 0, navigate to the CLI tab, and enter the following commands.

```
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
Router#enable  
Router#config terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#router rip  
Router(config-router)#no auto-summary //needed to use rip v2  
Router(config-router)#version 2 //using rip v2  
Router(config-router)#network 192.168.1.0  
Router(config-router)#network 192.168.3.0  
Router(config-router)#
```

Next, go to Router 1, navigate to the CLI tab, and enter the following commands.

```
Router#  
%SYS-5-CONFIG_I: Configured from console by console  
Router#enable  
Router#config terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#router rip  
Router(config-router)#no auto-summary //needed to use rip v2  
Router(config-router)#version 2 //using rip v2  
Router(config-router)#network 192.168.2.0  
Router(config-router)#network 192.168.3.0  
Router(config-router)#
```

Now RIP has been implemented on both routers and they can discover other networks through the RIP protocol.

Step 3: Verifying Network Connectivity

After configuring RIP on both routers, the routing tables should automatically update, allowing communication between the two networks. To verify connectivity, click on PC0, go to the Desktop tab, and open the Command Prompt. First, test connectivity by sending a ping to PC2. If the configuration is correct, you should receive reply messages, indicating successful communication between the networks. Next, to observe the path taken by the packets, use the `tracert 192.168.2.2` command. This command displays the route (hop by hop path) that packets take from PC0 to PC2. You should see the packet traveling from PC0 to Router 0, then to Router 1, and finally reaching PC2. This confirms that RIP is correctly exchanging routing information between the routers.



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Submission Guidelines (5 Marks):

- Submit a single PDF document containing all tasks and the CISCO packet file(s) to the GCR, **also ensure your file name is visible in all pictures and contains your roll number.**
- Ensure that all screenshots are clear and labeled appropriately.
- Include your name, roll number, and lab number inside the **first page** of the document.
- Name the submitted document as follows: SectionLetter_RollNumber_LabNumber, for example: **L_i231234_Lab1.**
- Name the CISCO files as follows: SectionLetter_RollNumber_TaskNumber, for example: **L_i231234_Lab1. (for multiple files add the task number at the end)**
- The unethical usage of Generative AI is strictly prohibited and will result in 0 marks.
- **In case a demo is conducted, there will be deductions based on your answers to the questions asked.**

NOTE: Screenshots where your name, roll number, section AND lab number are NOT visible, will automatically be given 0/not considered valid.

Sample screenshot:

